BIOMOLECULES

Biomolecules are the organic and inorganic molecules present in living organisms, essential for the structure and function of cells.

Chemical Composition Analysis

Methods →

⇒ Elemental analysis finds elements like C, H, O, N, S etc.

⇒ **Quantitative comparision** :- Living tissue v.s Earth's crust differs in relative abundance of elements (e.g. C, H, O much higher in organisms).

Types of Biomolecules

A.) **Based on Molecular weight** :- Low molecular weight (<1000 Da) → **Micromolecules**

Examples :- Amino acids, Sugars, nucleotides, fatty acids.

- **Macromolecules** :- Higher molecular weight (>10,000 Da)

Examples :- Proteins, Nucleic acids, polysaccharides, some lipids.

B.) **Based On Organic class**

Biomolecule	Building block	Examples	Function
Proteins	Amino acids	Enzymes, Collagen	Structure, catalyst, transport
Carbohydrates	Monosaccharides	Glucose, starch	Energy storage, structure
Lipids	fatty acids, glycerol	Triglyceride, lecithin	Energy storage, Membrane structure
Nucleic acids	Nucleotides	DNA, RNA	Genetic information

Primary & Secondary Metabolites

- **Primary Metabolites :-** Directly involved in normal growth, development, and reproduction (e.g.) amino acids, sugars)

- **Secondary Metabolites :-** Not essential for survival but beneficial (e.g.) alkaloids, flavonoids, antibiotics, pigments).

Biomacromolecules

⇒ Micromolecules :- Molecular weights 18 to ~800 Da. present in the acid-soluble pool.

⇒ Macromolecules :- Molecular weights 10,000 Da and above. Present in the acid-insoluble fraction.

⇒ Proteins, nucleic acids and polysaccharides are true macromolecules and polymeric substances.

A Comparison of Elements present in non-living and Living matter.

Element	% weight of Earth's crust	% weight of Human body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

A List of Representative inorganic Constituents of Living tissues

Component	Formula
Sodium	Na^+
Potassium	K^+
Calcium	Ca^{++}
Magnesium	Mg^{++}
water	H_2O
Compounds	Nacl, $CaCO_3$, PO_4^{3-}, SO_4^{--}

Average composition of cells

Component	% of total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

Examples of Secondary metabolites

Pigments	Carotenoids, Anthocyanins
Alkaloids	Morphine, codeine
Terpenoides	Monoterpenes, Diterpenes
Essential oils	Lemon grass oil
Toxins	Abrin, Ricin
Lectins	concanavalin A
Drugs	vinblastin, curcumin
Polymeric substances	Rubber, gums, cellulose

Lipids

→ They could be simple fatty acids.

→ A fatty acid has a carboxyl group attached to an R group.

→ The R group could be from 1 carbon to 19 carbons.

eg:- Palmitic acid has 16 carbons including carboxyl carbon.

$CH_3-(CH_2)_{14}-COOH$

eg:- Arachidonic acid has 20 carbon including the carboxyl carbon.

→ Fatty acid could be saturated or unsaturated

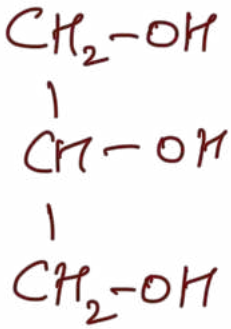
saturated → C-C

unsaturated → one or more C=C

→ Simple lipid is glycerol

↓

Trihydroxy propane



→ Many lipids have both fatty acids and glycerol

→ Fatty acid esterified with glycerol and form monoglyceride, diglyceride and triglyceride.

Types on the basis of melting point

(i) Oil → Oil have lower melting point and hence remain as oil in winter.

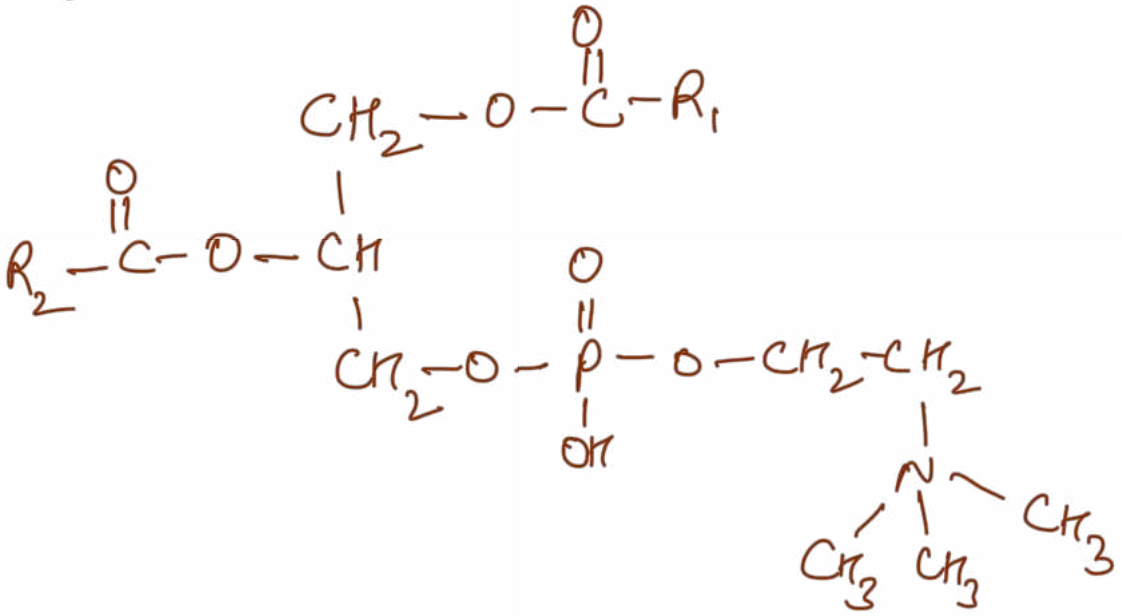
eg:- gingelly oil

(ii) Fat → Fats have higher melting point.

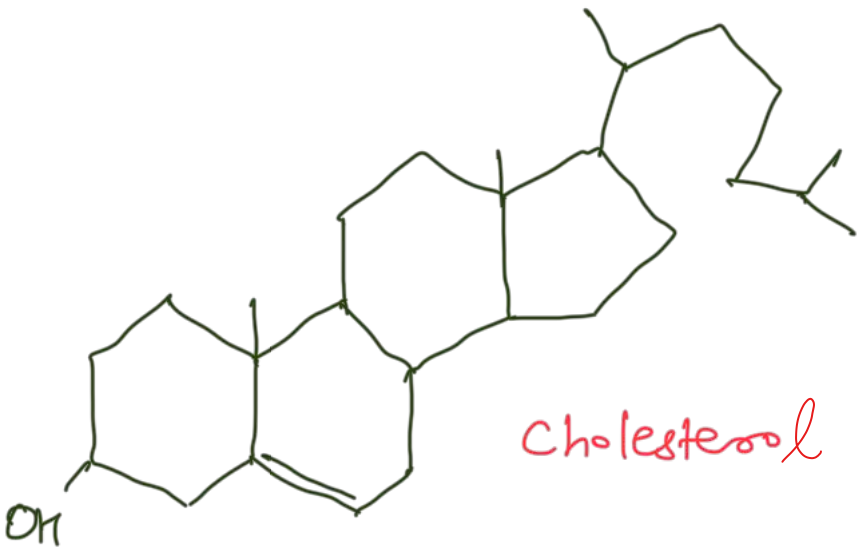
Phospholipid

→ Lipids having phosphorus or phosphorylated organic compounds in them.

Eg:- Lecithin



→ They are found in cell membrane.



⇒ Generally water-insoluble.

⇒ Molecular weights do not exceed 800 Da, so they are not strictly macromolecules.

⇒ However, they are found in the acid-insoluble fraction because they are arranged into cell membranes and other structures. When tissues are ground, these membrane break into water-insoluble vesicles that separate with the acid-insoluble pool.

⇒ Cytoplasmic Composition :-

- The acid-soluble pool roughly represents the cytoplasmic Composition.
- Macromolecules from cytoplasm and organelles from the acid-insoluble fraction.

Proteins :-

⇒ Polypeptides; linear chains of amino acids linked by peptide bonds.

⇒ Proteins are heteropolymers because they are made of 20 different types of amino acids (monomers). (A homopolymer has only one type of monomer repeating).

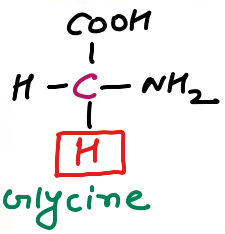
• Amino acids :-

⇒ Essential Amino Acids :- Must be supplied through diet.

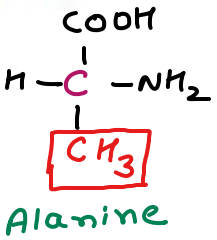
⇒ Non-Essential Amino Acids :- Can be synthesized by the body.

⇒ Amino acids contain an amino group and an acidic (carboxyl) group attached to the same alpha-carbon, alongwith a hydrogen and a variable R-group.

⇒ Amino acids exist in zwitterionic in solutions of different pH.



Glycine



Alanine

Functions of Proteins

⇒ Transport nutrients across cell membranes (e.g.) GLUT 4)

⇒ Fight infectious organisms (e.g.) Antibodies)

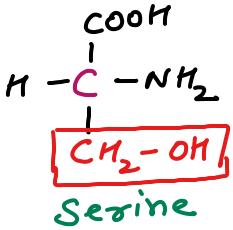
⇒ Hormones (e.g.) Insulin)

⇒ Enzymes (e.g.) Trypsin)

⇒ Receptors (for sensory reception)

⇒ Intercellular ground substance (e.g.) Collagen)

⇒ Structural proteins.



Serine

- **Most Abundant Proteins :-**

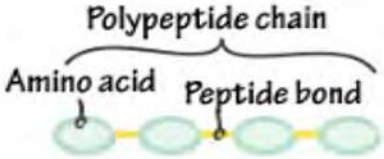
→ Collagen : Most abundant protein in the animal world.

→ RuBisCo : (Ribulose bisphosphate carboxylase-Oxygenase) Most abundant protein in the whole biosphere.

Structure of Proteins

1.) Primary Structure :-

⇒ The specific sequence of amino acids in a polypeptide chain.



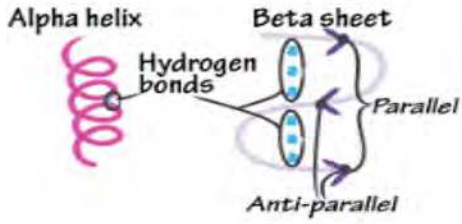
⇒ The left end is the N-terminal amino acid (first), and the right end is the C-terminal amino acid (last).

2.) Secondary Structure :-

⇒ folding of the polypeptide chain into specific recurring structures.

i.) **Alpha-Helix :-** A right-handed coiled structure, similar to a revolving staircase.

ii.) **Beta-pleated sheet :-** Another common folded structure.

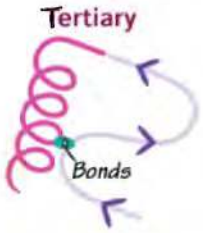


3.) Tertiary Structure :-

⇒ The overall three-dimensional folding of the entire polypeptide chain, forming a hollow woolen ball-like structure.

⇒ Maintained by various bonds (e.g.) hydrogen bonds, disulphide bonds).

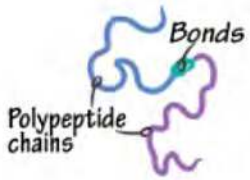
⇒ Absolutely necessary for the biological activities of proteins.



4.) Quaternary Structure :-

⇒ The arrangement of multiple polypeptide subunits (if a protein has more than one) with respect to each other.

⇒ Example :- Adult human haemoglobin (Hb) consists of 4 subunits (two alpha two beta types).



Polysaccharides

⇒ Long chains of sugars (monosaccharides).

⇒ Examples :-

1.) Cellulose :- A Homopolymer of only one type of monosaccharide, glucose. Found in plant cell walls, paper, and cotton fiber.

2.) **Starch** :- A varient of cellulose, serving as a storehouse of energy in plant tissues. forms helical secondary structures that can hold I_2 molecules, turning blue.

3.) **Glycogen** :- An energy storehouse in animals. Has reducing and non-reducing ends and branches.

4.) **Inulin** :- A polymer of fructose.

Complex Polysaccharides :- Have amino-sugars and chemically modified sugars. as building blocks (e.g.) glucosamine, N-acetyl galactosamine).

- **Chitin** :- A complex polysaccharide found in the exoskeleton of arthropods. Mostly homopolymers.

function → Components of cell walls in plants and fungi, exoskeleton of Arthropods, Storage forms of energy.

Nucleic Acids

⇒ Polynucleotides; true macromolecules found in the acid-insoluble fraction.

Building block : Nucleotide

Components of a Nucleotide :-

1. Heterocyclic Compound (Nitrogenous Base)
   - Purines :- Adenine (A), Guanine (G), (Double ring structure)
   - Pyrimidines :- Cytosine (C), Uracil (U), Thymine (T). (Single-ring structures).

Adenine



Purine

Uracil

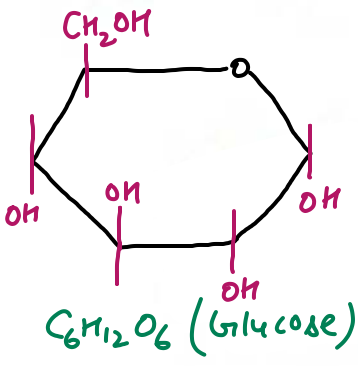


Pyrimidine

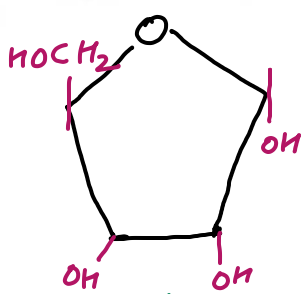
Monosaccharide (Sugar) :

⇒ Ribose :- found in Ribonucleic Acid (RNA).

⇒ 2'- deoxyribose :- found in de-oxy ribonucleic acid (DNA)



$C_6H_{12}O_6$ (Glucose)



$C_5H_{10}O_5$ (Ribose)

Chemical Reaction

→ A physical change refers to a change in shape without breaking of bonds.

→ In chemical reaction, bonds are broken and new bonds are formed.

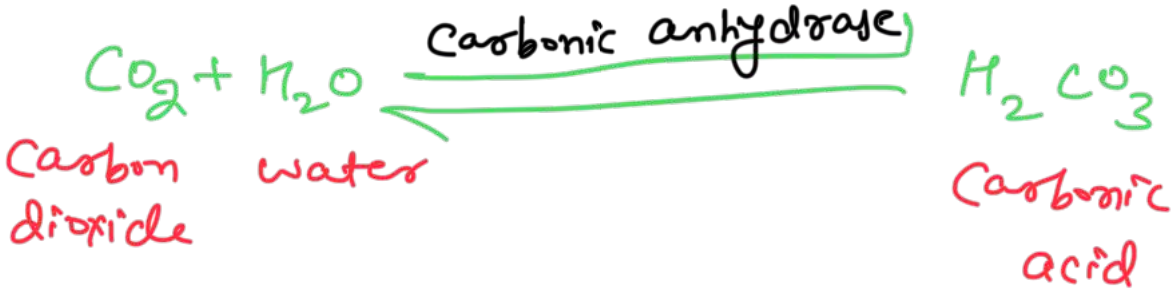
eg:- $Ba(OH)_2 + H_2SO_4 \longrightarrow BaSO_4 + 2H_2O$

→ Rate of a physical or chemical process refers to the amount of product formed per unit time.

$$\text{Rate} = \frac{\delta P}{\delta t}$$

→ A general rule of thumb is that rate doubles or decreases by half for every 10°C change in either direction.

→ Catalysed reaction proceeds at rates vastly higher than that of uncatalysed ones.



→ In the absence of any enzyme this reaction is very slow with about 200 molecules of H_2CO_3 being formed in an hour.

→ By using the enzyme present within the cytoplasm called carbonic anhydrase, the reaction speeds dramatically with about 600,000 molecules being formed every second.

→ The enzyme has accelerated the rate of reaction by about 10 millions times.

Glucose ⟶ 2 Pyruvic acid

$C_6H_{12}O_6 + O_2 \longrightarrow 2C_3H_4O_3 + 2H_2O$

It is a multistep reaction, where different types of enzymes are involved at each steps.

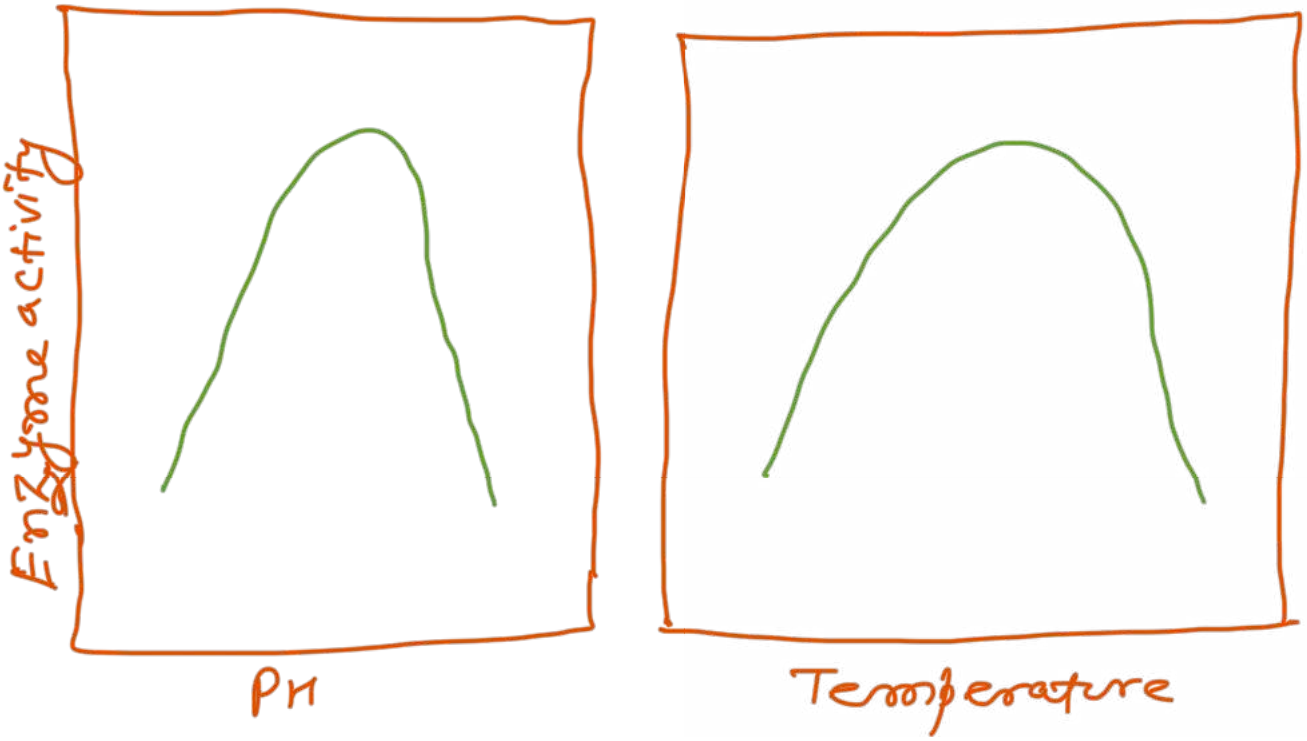
Pyruvic acid

- Aerobic Condition → Lactic acid
- Anaerobic Condition → Pyruvic acid
- Yeast → Ethanol (by fermentation)

Types of reaction

1. Endothermic reaction → The reaction in which energy of product is more than reactant.
2. Exothermic reaction → The reaction in which energy of product is less than reactant.

Factors affecting enzyme activity



Classification and nomenclature of Enzyme

→ Enzymes are divided into 6 classes each with 4-13 subclasses and named according by a four digit number.

① oxidoreductase/dehydrogenases

→ Enzyme which catalyse oxidoreduction between a pair of substrates.

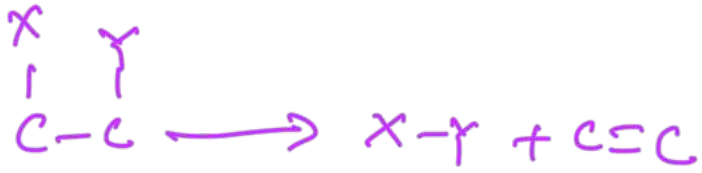
S reduced + S' oxidised ⟶ S' reduced + S oxidised

② Transferases → Enzymes catalysing a transfer of a group (other than hydrogen) between a pair of substrates.

S-G + S' ⟶ S'-G + S

③ Hydrolases → Enzymes catalysing hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds.

④ Lyases → Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds.



⑤ Isomerases → Enzymes catalysing inter-conversion of optical, geometrical or positional isomers.

⑥ Ligases → Enzymes catalysing the linking together of 2 compounds.

eg:— Enzymes which catalyse joining of C–O, C–S, C–N, P–O etc bonds.

Enzyme

⇒ **Nature** :- • Almost all enzymes are proteins
• Some Nucleic acids (Ribozymes) also act as enzymes

⇒ **function** :- Biological catalysts that increase reaction rates without being consumed.

⇒ **Active Site** :- Cervices / pocket where Substrate binds.

⇒ **Catalytic Cycle** :-

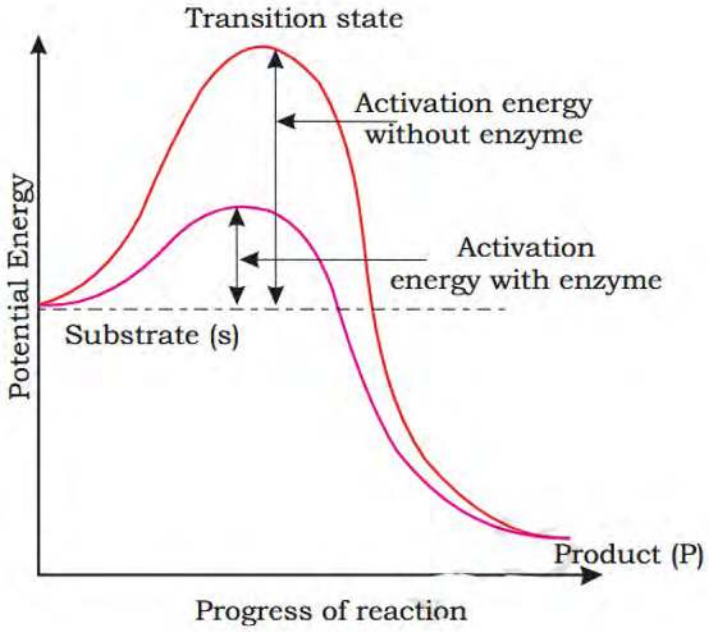
1.) Substrate (S) binds to enzyme (E) active site → ES complex

2.) Enzyme changes shape, fitting substrate tightly (induced fit)

3.) Chemical bonds of Substrate broken / New ones formed → EP-complex

4.) Product (P) released, enzyme free to bind new substrate.

Activation Energy :- Enzyme lower the activation energy required for a reaction to proceed.



factors affecting Enzyme Activity :-

1.) Temperature :- Optimum temperature for highest activity.

→ Activity declines above / below optimum

→ High temp denatures enzymes.

** ⇒ Thermophilic Organisms have stable enzymes at high temp (80-90°C)

2.) **pH** :- Optimum pH for highest activity.
Activity declines above/below optimum.

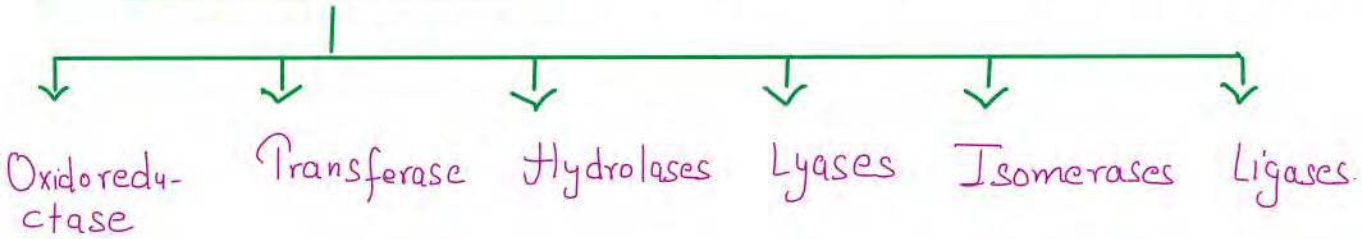
3.) **Substrate Concentration** :- Velocity (V) increases with [S] until V_{max} is reached (enzyme Saturation).

4.) **Inhibitors** :- Chemicals that shut off enzyme activity.

Competitive Inhibitor :-

Resembles Substrate, Compete for active site
(e.g.) Malonate inhibits Succinic dehydrogenase)

⇒ **Classification** :- 6 classes based on type of reaction



⇒ **Cofactors** :- Non-protein Constituents essential for catalytic activity.

⇒ **Apoenzyme** :- Protein portion of enzyme.

⇒ **Prosthetic Groups** :- Organic Compounds, tightly bound (e.g.) haem in peroxidase / catalase).

⇒ **Co-enzymes** :- Organic Compounds, transiently associated (e.g.) NAD, NADP contain niacin).

⇒ **Metal Ions** :- form Coordination bonds with active site and Substrate

(e.g.) Zinc in Carboxypeptidase).